

## Machine Learning – HW#1

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### General notes:

- It is **recommended** that you work alone to learn the material. However, groups of size 2 are allowed.
- Read and understand the questions, and answer accordingly. Be sure to answer what is asked. When in doubt, you can provide more details.
- Some of the questions require material not covered in class, or will be covered in future lectures. This is intentional.
- **Warning:** do not copy other student work. This includes code similarity. Do not sit and solve assignments together, unless it is your group partner.

### Submission document format guidelines:

- Submit pdf document to Gradescope.
- You can use word, latex, or any other editor.
- No need for cover page. Provide your names and IDs in the header.
- Answer each Question on a new page. No need to copy the question, just put Question # at the beginning, then a), b), etc. for each sub-question.
- When submitting to Gradescope, you will be given the chance to map pages to questions.
- For groups, one student submits and adds their teammate during submission on Gradescope.

### **Question 1 [6 pts]**

Suppose  $A, B, C, D$  are **binary** random variables. For each of the following specify how many entries/parameter?

The first one is done as an example.

**a)**  $P(A \mid C)$

2 entries for  $P(A \mid C = \text{true})$  that sums to 1  
2 entries for  $P(A \mid C = \text{false})$  that sums to 1  
Total 4 entries.

**b)**  $P(A, B, C, D)$

**c)**  $P(A, D \mid B = \text{true}, C = \text{true})$

**d)**  $P(B \mid A = \text{true}, C, D)$

## Question 2 [7 pts]

Consider the Dental Patient environment with variables (toothache, catch, cavity) with the full-joint table:

|                      | <i>toothache</i> |                     | $\neg$ <i>toothache</i> |                     |
|----------------------|------------------|---------------------|-------------------------|---------------------|
|                      | <i>catch</i>     | $\neg$ <i>catch</i> | <i>catch</i>            | $\neg$ <i>catch</i> |
| <i>cavity</i>        | <b>.108</b>      | <b>.012</b>         | <b>.072</b>             | <b>.008</b>         |
| $\neg$ <i>cavity</i> | <b>.016</b>      | <b>.064</b>         | <b>.144</b>             | <b>.576</b>         |

Answer the following probabilities: (show your work)

a)  $P(\neg \text{toothache}, \text{catch})$

b)  $P(\text{toothache} \mid \text{cavity})$

c)  $P(\neg \text{cavity} \mid \text{toothache} \vee \text{catch})$

### **Question 3 [20 pts]**

You are given the following information:

$A \sim \text{Uniform}(10, 50)$  where  $A \in \mathbb{R}$

$B \sim \text{Bernoulli}(p = 0.35)$  where  $B \in \{\text{yes}, \text{no}\}$  and  $p$  is yes probability

$C \sim \text{Cat}([0.2, 0.4, 0.4])$  where  $C \in \{\text{circle}, \text{triangle}, \text{square}\}$

$D \sim \text{Normal}(\mu = 100, \sigma = 30)$  note  $\sigma$  is **standard deviation**.

Answer the following questions using numpy and seaborn packages.

a) For each of these variables:

- Sample  $m = 50$  using numpy package.
- Plot the histogram using seaborn package.  
<https://seaborn.pydata.org/generated/seaborn.histplot.html>
- Comment on the shape of the histogram. Is it as expected? How to improve the shape?

b) Suppose that you have another variable  $E \sim \text{Normal}(\mu = 20, \sigma = 20)$ .

- Sample  $m = 200$  for the two variables  $D$  and  $E$ .
- Make a 2d plot of these  $(D, E)$  samples.
- What can you say about the [covariance](#) between  $D$  and  $E$ ? See also [link](#).

**Support your answer by showing snapshots of code and visuals (use snipping tool).**

#### **Question 4 [10 pts]**

Suppose we have  $n$  independent and identically distributed samples  $x_1, x_2, \dots, x_n$  where  $x_i \sim \text{Normal}(\mu, \sigma^2)$

Use MLE formulation and show that the MLE estimate  $\hat{\mu}$  for  $\mu$  is the sample average.

Hint1: Although we do not know  $\sigma$ , at some point you can drop it as a constant.  
 $\text{argmax } c f(x) = \text{argmax } f(x)$  where  $c$  is a constant

Hint2: use the fact that finding max of a function is equivalent to finding min of the negative of that function.  $\text{argmax } f(x) = \text{argmin } -f(x)$

Hint3: how to find the minimum of a function?

Show clear derivation steps.

### **Question 5 [20 pts]**

You are given a csv file `q5_data.csv` that contains 4 features:  $F1, F2, F3, F4$ . Your task is to analyze this data.

a) For *each* feature:

- Determine the most appropriate common scalar distribution it came from.
- After determining the scalar distribution, estimate its parameters (MLE estimate).

b)  $F3$  and  $F4$  are related. Determine this relationship. (Hint: make a 2D plot of  $(f3, f4)$  points)

Note: you **should** use python packages to **support** your answer. Pandas to read the csv file and seaborn to visualize. A helpful visualization is

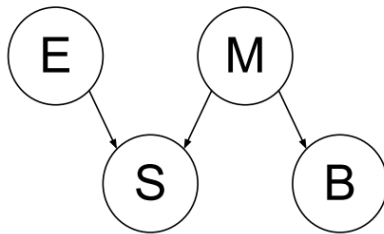
<https://seaborn.pydata.org/generated/seaborn.histplot.html>. Show snapshots using snipping tool.

### Question 6 [9 pts]

Consider the following Bayes' Net.

| $P(E)$ |     |
|--------|-----|
| $+e$   | 0.4 |
| $-e$   | 0.6 |

| $P(S   E, M)$ |      |      |     |
|---------------|------|------|-----|
| $+e$          | $+m$ | $+s$ | 1.0 |
| $+e$          | $+m$ | $-s$ | 0.0 |
| $+e$          | $-m$ | $+s$ | 0.8 |
| $+e$          | $-m$ | $-s$ | 0.2 |
| $-e$          | $+m$ | $+s$ | 0.3 |
| $-e$          | $+m$ | $-s$ | 0.7 |
| $-e$          | $-m$ | $+s$ | 0.1 |
| $-e$          | $-m$ | $-s$ | 0.9 |



| $P(M)$ |     |
|--------|-----|
| $+m$   | 0.1 |
| $-m$   | 0.9 |

| $P(B   M)$ |      |     |
|------------|------|-----|
| $+m$       | $+b$ | 1.0 |
| $+m$       | $-b$ | 0.0 |
| $-m$       | $+b$ | 0.1 |
| $-m$       | $-b$ | 0.9 |

a). What is  $P(-e, -s, -m, -b)$ ?

b) What is  $P(+b)$ ?

c) What is  $P(+m | +b)$ ?

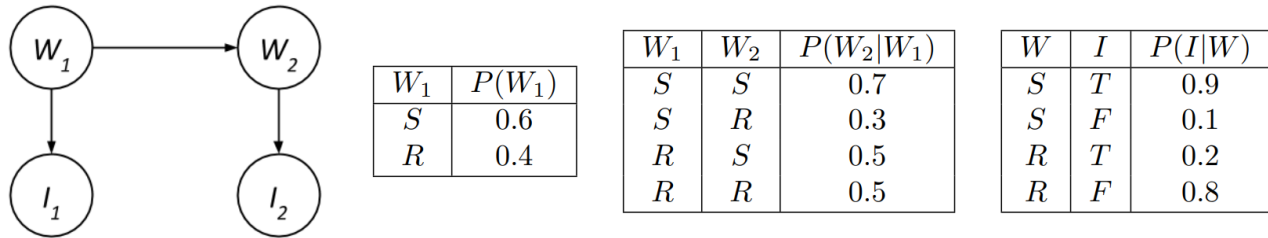
d) What is  $P(+m | +s, +b, +e)$ ?

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### Question 7 [15 pts]

Suppose you are given the following Bayes' Net:



For the following questions use [pgmpy](#) package.

**a)** Construct this Bayesian network in [pgmpy](#). Show snapshots of your code, provide commentary, and plot the structure using graphviz (see example [here](#)).

Note that you should take care when constructing the CPT tables. Be sure to see examples to get an idea.

**b)** Using the BN model, use rejection sampling with 20 samples to approximate  $P(W_2 = R \mid I_1 = T, I_2 = F)$

Note: you should perform rejection sampling first (see [here](#)), then approximate inference with those samples (see [here](#)). Show code snapshots.

**c)** Use likelihood weighting with 20 samples to estimate  $P(W_2 = R \mid I_1 = T, I_2 = F)$ . Show code snapshots.

### **Question 8 [16 pts]**

Consider a supervised learning problem where the data has 3 features and 1 continuous output label. Furthermore, consider that we use the following model:

$x_{i1} \sim \text{Uniform}(0, 10)$  where  $x_{i1}$  follows continuous uniform distribution.

$$x_{i2} = 2x_{i1} + 5$$

$x_{i3} \sim \text{Bernoulli}(p = 0.2)$  where  $x_{i3} \in \{0, 1\}$  and  $p$  is probability of 1

$$y_i | x_i \sim \text{Normal}(\mu = x_{i3}x_{i1} + (1 - x_{i3})x_{i2}, \sigma = 2)$$

- a) Draw the sampling graph for an instance of this problem. Arrows should indicate order of sampling.
- b) What can you say about the distribution of  $x_{i2}$ ?
- c) Is this model generative or discriminative? Give brief justification for your answer.
- d) What can you say about the distribution of  $y_i | x_i$ ? In particular, what is the expected mean of  $y_i | x_i$  when  $x_{i3} = 0$  versus when  $x_{i3} = 1$ .